

Lecture 15

ANGULAR MOMENTUM, MOMENT OF A FORCE AND PRINCIPLE OF ANGULAR IMPULSE AND MOMENTUM

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ANGULAR MOMENTUM, MOMENT OF A FORCE AND PRINCIPLE OF ANGULAR IMPULSE AND MOMENTUM

Today's Objectives:

Students will be able to:

1. Determine the angular momentum of a particle and apply the principle of angular impulse & momentum.
2. Use conservation of angular momentum to solve problems.



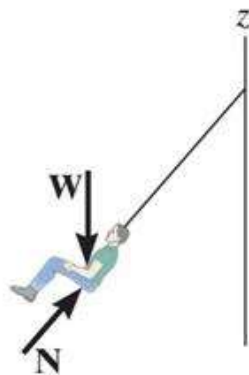
In-Class Activities:

- Reading Quiz
- Applications
- Angular Momentum
- Angular Impulse and Momentum Principle
- Conservation of Angular Momentum
- Concept Quiz
- Group Problem Solving
- Attention Quiz

READING QUIZ

1. Select the correct expression for the angular momentum of a particle about a point.
A) $\mathbf{r} \times \mathbf{v}$ B) $\mathbf{r} \times (m \mathbf{v})$
C) $\mathbf{v} \times \mathbf{r}$ D) $(m \mathbf{v}) \times \mathbf{r}$
2. The sum of the moments of all external forces acting on a particle is equal to
A) angular momentum of the particle.
B) linear momentum of the particle.
C) time rate of change of angular momentum.
D) time rate of change of linear momentum.

APPLICATIONS

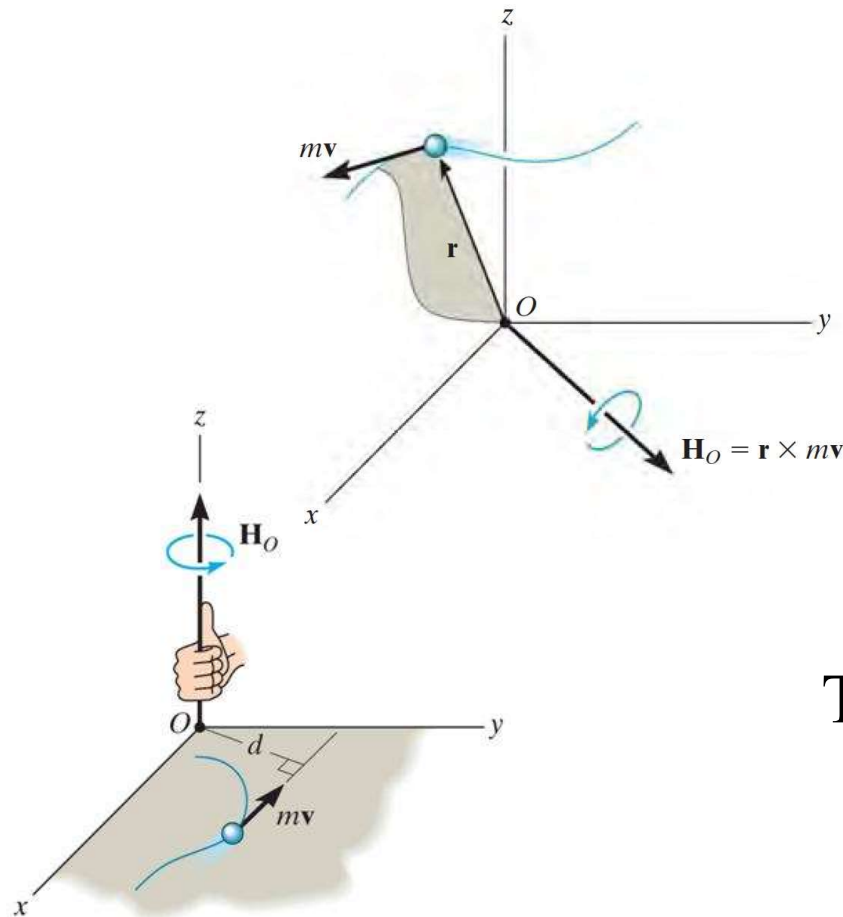


The passengers on the amusement-park ride experience conservation of angular momentum about the axis of rotation (the z-axis). As shown on the free body diagram, the line of action of the normal force, N , passes through the z-axis and the weight's line of action is parallel to it. Therefore, the sum of moments of these two forces about the z-axis is zero.

If the passenger moves away from the z-axis, will his speed increase or decrease? Why?

ANGULAR MOMENTUM (Section 15.5)

The **angular momentum** of a particle about point O is defined as the “moment” of the particle’s **linear momentum** about O.



$$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & r_z \\ mv_x & mv_y & mv_z \end{vmatrix}$$

The magnitude of $\mathbf{H}_O$ is $(H_O)_z = (d)(mv)$

RELATIONSHIP BETWEEN MOMENT OF A FORCE AND ANGULAR MOMENTUM (Section 15.6)

The resultant force acting on the particle is equal to the time rate of change of the particle's linear momentum. Showing the time derivative using the familiar “dot” notation results in the equation

$$\sum \mathbf{F} = \dot{\mathbf{L}} = m\dot{\mathbf{v}}$$

We can prove that the resultant moment acting on the particle about point O is equal to the time rate of change of the particle's angular momentum about point O or

$$\dot{\mathbf{H}}_O = \frac{d}{dt}(\mathbf{r} \times m\mathbf{v}) = \dot{\mathbf{r}} \times m\mathbf{v} + \mathbf{r} \times m\dot{\mathbf{v}}$$

$$\sum \mathbf{M}_O = \mathbf{r} \times \mathbf{F} = \dot{\mathbf{H}}_O$$

PRINCIPLE OF ANGULAR IMPULSE AND MOMENTUM (Section 15.7)

Considering the relationship between moment and time rate of change of angular momentum

$$\sum \mathbf{M}_o = \dot{\mathbf{H}}_o = d\mathbf{H}_o/dt$$

By integrating between the time interval t_1 to t_2

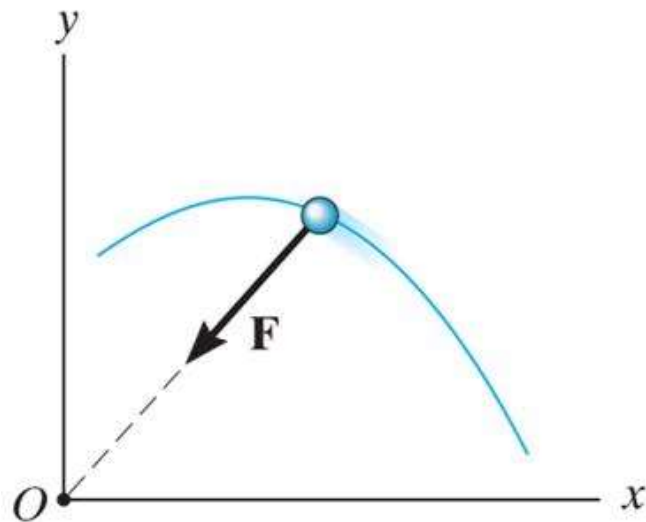
$$\sum \int_{t_1}^{t_2} \mathbf{M}_o dt = (\mathbf{H}_o)_2 - (\mathbf{H}_o)_1 \quad \text{or} \quad (\mathbf{H}_o)_1 + \sum \int_{t_1}^{t_2} \mathbf{M}_o dt = (\mathbf{H}_o)_2$$

This equation is referred to as the **principle of angular impulse and momentum**. The second term on the left side, $\sum \int M_o dt$, is called the **angular impulse**. In cases of 2D motion, it can be applied as a scalar equation using components about the z-axis.

CONSERVATION OF ANGULAR MOMENTUM

When the sum of angular impulses acting on a particle or a system of particles is zero during the time t_1 to t_2 , the angular momentum is conserved. Thus,

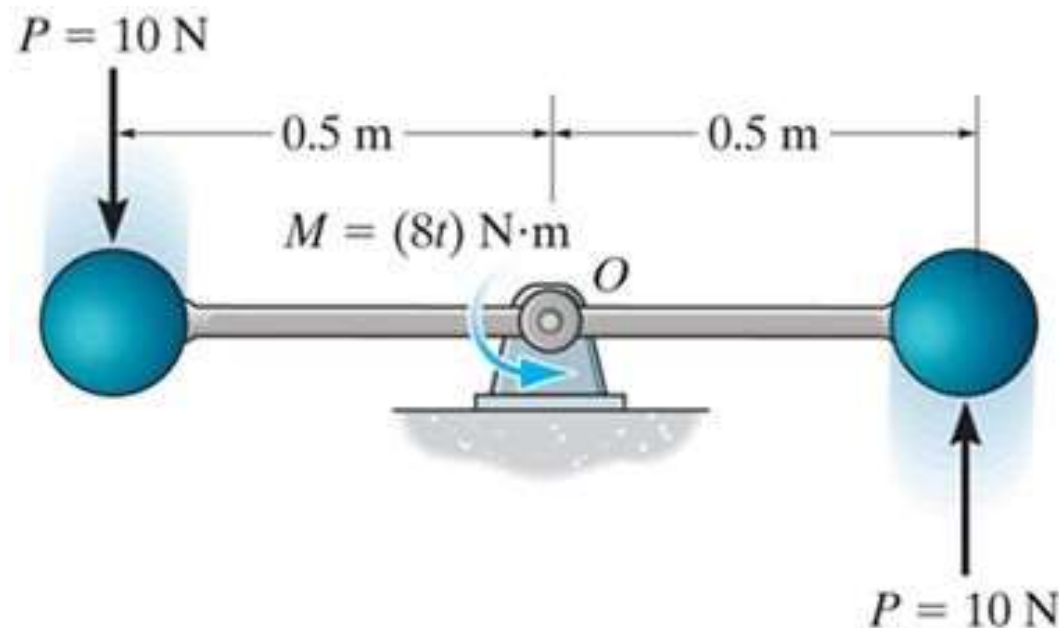
$$(\mathbf{H}_O)_1 = (\mathbf{H}_O)_2$$



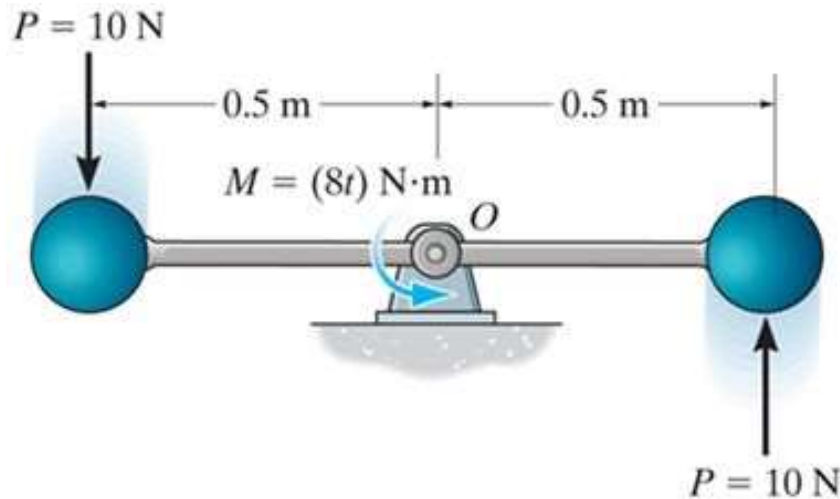
An example of this condition occurs when a particle is subjected only to a central force. In the figure, the force $\mathbf{F}$ is always directed toward point O. Thus, the angular impulse of $\mathbf{F}$ about O is always zero, and angular momentum of the particle about O is conserved.

EXAMPLE

Two identical 10-kg spheres are attached to the light rigid rod, which rotates in the horizontal plane centred at pin O. If the spheres are subjected to tangential forces of $P = 10 \text{ N}$, and the rod is subjected to a couple moment $M = (8t) \text{ N}\cdot\text{m}$, where t is in seconds, determine the speed of the spheres at the instant $t = 4 \text{ s}$. The system starts from rest. Neglect the size of the spheres. **(Problem No. 15-24)**



EXAMPLE



Given: Two identical 10-kg spheres are attached to the rod, which rotates in the plane of the page. The spheres are subjected to tangential forces of $P = 10 \text{ N}$, and the rod is subjected to a couple moment $M = (8 t) \text{ N}\cdot\text{m}$, where t is in seconds.

Find: The speed of the spheres at $t = 4 \text{ s}$, if the system starts from rest.

Plan: Apply the principles of conservation of energy and conservation of angular momentum to the system.

EXAMPLE (continued)

Solution:

Principle of angular impulse and momentum

$$\Sigma(\mathbf{H}_O)_1 + \Sigma \int_{t_1}^{t_2} \mathbf{M}_O dt = \Sigma(\mathbf{H}_O)_2$$

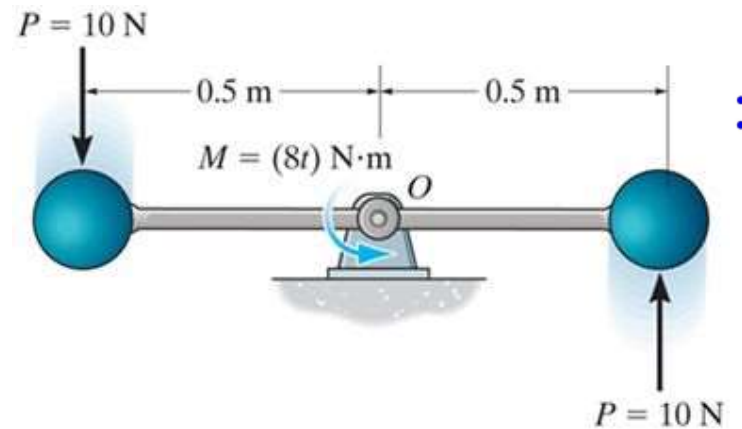
The above equation about the axis of rotation (z-axis) through O can be written as

$$0 + \int_0^4 8t dt + \int_0^4 [2(10)(0.5)] dt = 2 [10 v (0.5)]$$

$$\Rightarrow 4(4)^2 + 2(5)4 = 10 v$$

$$\Rightarrow 104 = 10 v$$

$$v = \underline{10.4 \text{ m/s}}$$

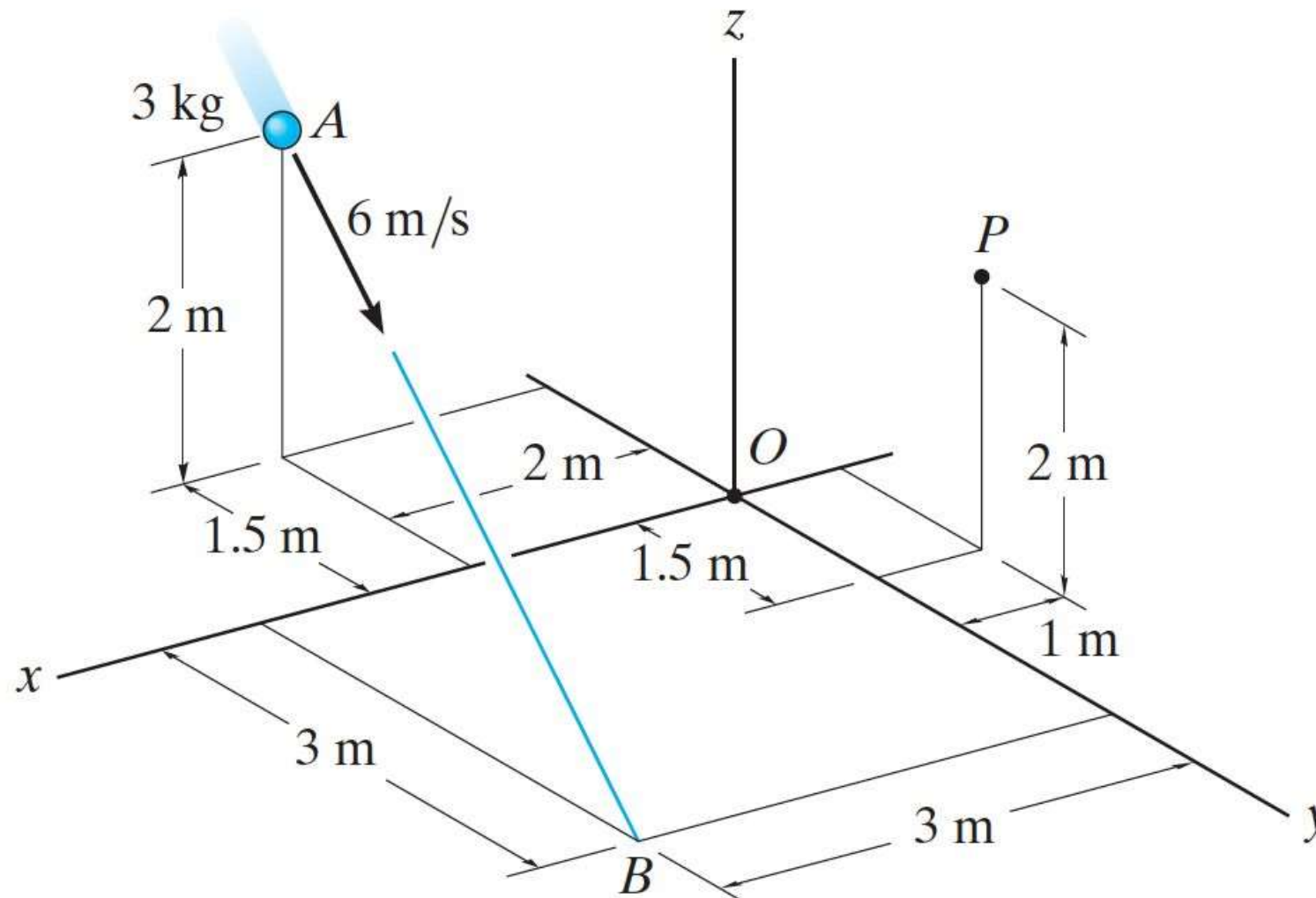


CONCEPT QUIZ

1. If a particle moves in the $x - y$ plane, its angular momentum vector is in the
 - A) x direction.
 - B) y direction.
 - C) z direction.
 - D) $x - y$ direction.
2. If there are no external impulses acting on a particle
 - A) only linear momentum is conserved.
 - B) only angular momentum is conserved.
 - C) both linear momentum and angular momentum are conserved.
 - D) neither linear momentum nor angular momentum are conserved.

GROUP PROBLEM SOLVING

Determine the angular momentum H_P of the 3-kg particle about point P. **(Problem No. 15-99)**



GROUP PROBLEM SOLVING

Position and Velocity Vectors. The coordinates of points A , B and P are $A(2, -1.5, 2)$ m, $B(3, 3, 0)$ m and $P(-1, 1.5, 2)$ m.

$$\mathbf{r}_{PA} = [2 - (-1)]\mathbf{i} + (-1.5 - 1.5)\mathbf{j} + (2 - 2)\mathbf{k} = \{3\mathbf{i} - 3\mathbf{j}\} \text{ m}$$

$$\mathbf{r}_{PB} = [3 - (-1)]\mathbf{i} + (3 - 1.5)\mathbf{j} + (0 - 2)\mathbf{k} = \{4\mathbf{i} + 1.5\mathbf{j} - 2\mathbf{k}\} \text{ m}$$

$$\mathbf{V}_A = v_A \left(\frac{\mathbf{r}_{AB}}{r_{AB}} \right) = 6 \left[\frac{(3 - 2)\mathbf{i} + [3 - (-1.5)]\mathbf{j} + (0 - 2)\mathbf{k}}{\sqrt{(3 - 2)^2 + [3 - (-1.5)]^2 + (0 - 2)^2}} \right]$$

$$= \left\{ \frac{6}{\sqrt{25.25}}\mathbf{i} + \frac{27}{\sqrt{25.25}}\mathbf{j} - \frac{12}{\sqrt{25.25}}\mathbf{k} \right\} \text{ m/s}$$

GROUP PROBLEM SOLVING

$$\mathbf{H}_p = \mathbf{r}_{pA} \times m\mathbf{V}_A$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -3 & 0 \\ 3\left(\frac{6}{\sqrt{25.25}}\right) & 3\left(\frac{27}{\sqrt{25.25}}\right) & 3\left(-\frac{12}{\sqrt{25.25}}\right) \end{vmatrix}$$

$$= \{21.4928\mathbf{i} + 21.4928\mathbf{j} + 59.1052\mathbf{k}\} \text{ kg} \cdot \text{m}^2/\text{s}$$

$$= \{21.5\mathbf{i} + 21.5\mathbf{j} + 59.1\mathbf{k}\} \text{ kg} \cdot \text{m}^2/\text{s}$$

Ans.

$$\mathbf{H}_p = \mathbf{r}_{pB} \times m\mathbf{V}_A$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 1.5 & -2 \\ 3\left(\frac{6}{\sqrt{25.25}}\right) & 3\left(\frac{27}{\sqrt{25.25}}\right) & 3\left(-\frac{12}{\sqrt{25.25}}\right) \end{vmatrix}$$

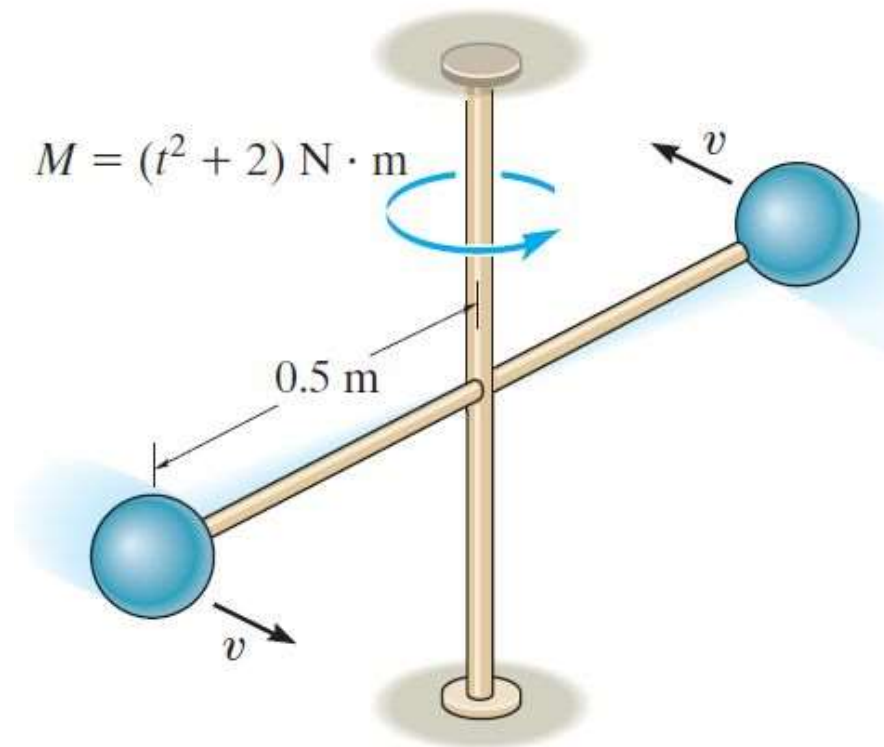
$$= \{21.4928\mathbf{i} + 21.4928\mathbf{j} + 59.1052\mathbf{k}\} \text{ kg} \cdot \text{m}^2/\text{s}$$

$$= \{21.5\mathbf{i} + 21.5\mathbf{j} + 59.1\mathbf{k}\} \text{ kg} \cdot \text{m}^2/\text{s}$$

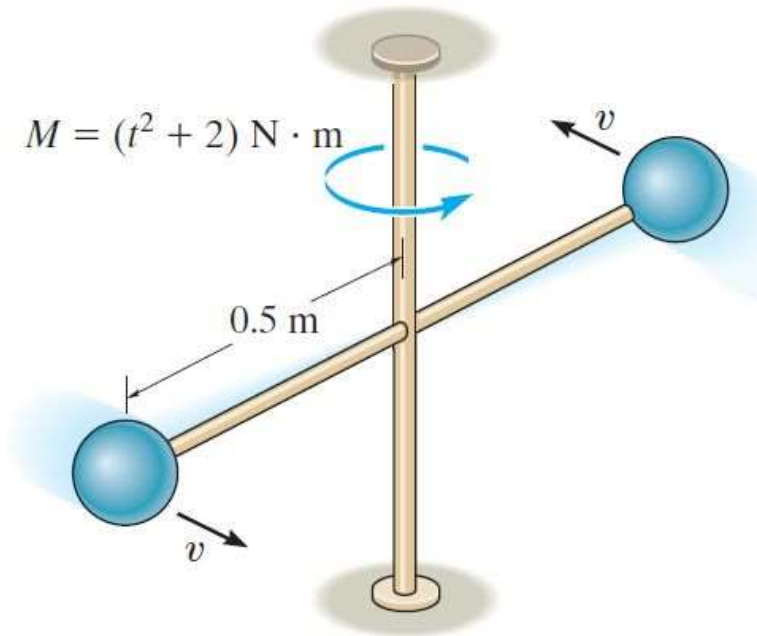
Ans.

GROUP PROBLEM SOLVING

Each ball has a negligible size and a mass of 10 kg and is attached to the end of a rod whose mass may be neglected. If the rod is subjected to a torque $M = (t^2 + 2) \text{ N}\cdot\text{m}$, where t is in seconds, determine the speed of each ball when $t = 3 \text{ s}$. Each ball has a speed $v = 2 \text{ m/s}$ when $t = 0$. **(Problem No. 15-100)**



GROUP PROBLEM SOLVING



Given: The two 10 kg balls are attached to the end of a rod of negligible weight. A torque acts on the rod as shown, $M = (t^2 + 2) \text{ N} \cdot \text{m}$.

Find: The velocity of each ball after 3 seconds, if each ball has a speed $v = 2 \text{ m/s}$ when $t = 0$.

Plan: Apply the principle of angular impulse and momentum about the axis of rotation (z-axis).

GROUP PROBLEM SOLVING (continued)

Solution:

Angular momentum: $H_Z = r \times m\mathbf{v}$ reduces to a scalar equation.

$$(H_Z)_1 = 2 \times \{0.5 (10) 2\} = 20 \text{ (kg}\cdot\text{m}^2/\text{s)} \text{ and}$$

$$(H_Z)_2 = 2 \times \{0.5 (10) v\} = 10 v \text{ (kg}\cdot\text{m}^2/\text{s)}$$

Angular impulse:

$$\int_{t_1}^{t_2} M dt = \int_{t_1}^{t_2} (t^2 + 2) dt = \left[(1/3) t^3 + 2 t \right]_0^3 = 15 \text{ N}\cdot\text{m}\cdot\text{s}$$

Apply the principle of angular impulse and momentum.

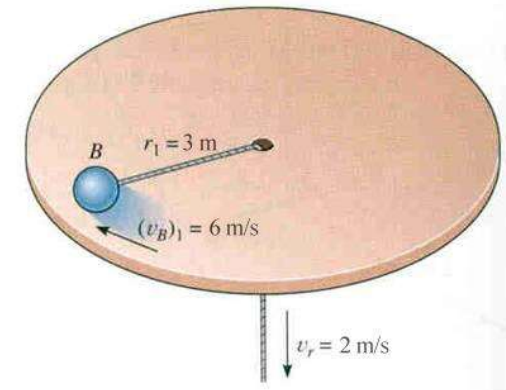
$$(\mathbf{H}_o)_1 + \sum \int_{t_1}^{t_2} \mathbf{M}_o dt = (\mathbf{H}_o)_2$$

$$20 + 15 = 10 v \Rightarrow v = \underline{3.5 \text{ m/s}}$$

ATTENTION QUIZ

1. A ball is traveling on a smooth surface in a 3-m radius circle with a speed of 6 m/s. If the attached cord is pulled down with a constant speed of 2 m/s, which of the following principles can be applied to solve for the velocity of the ball when $r = 2$ m?

- A) Conservation of energy
- B) Conservation of angular momentum
- C) Conservation of linear momentum
- D) Conservation of mass



2. If a particle moves in the $z - y$ plane, its angular momentum vector is in the
- A) x direction. B) y direction.
 - C) z direction. D) $z - y$ direction.

End of the Lecture
Let The Learning Continue